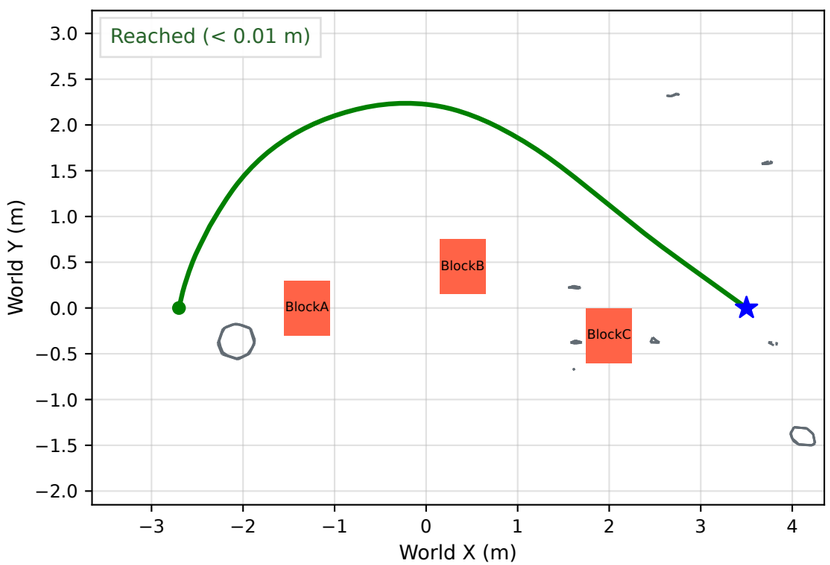
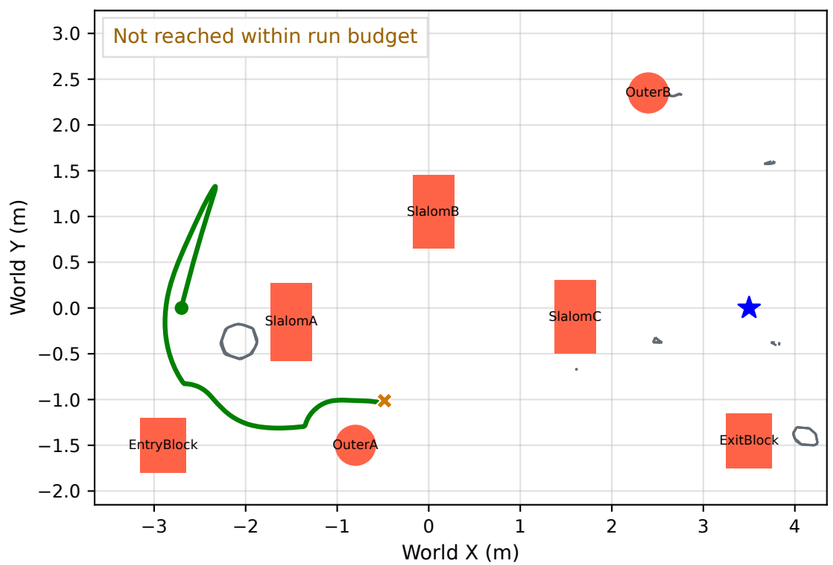
ROV obstacle-avoidance routes across five pool scenarios

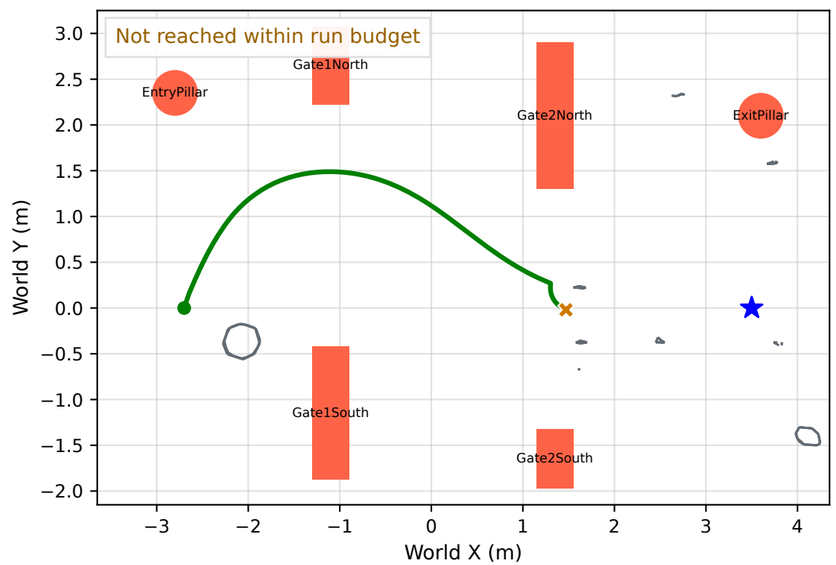
Baseline | goal error: 0.0036 m



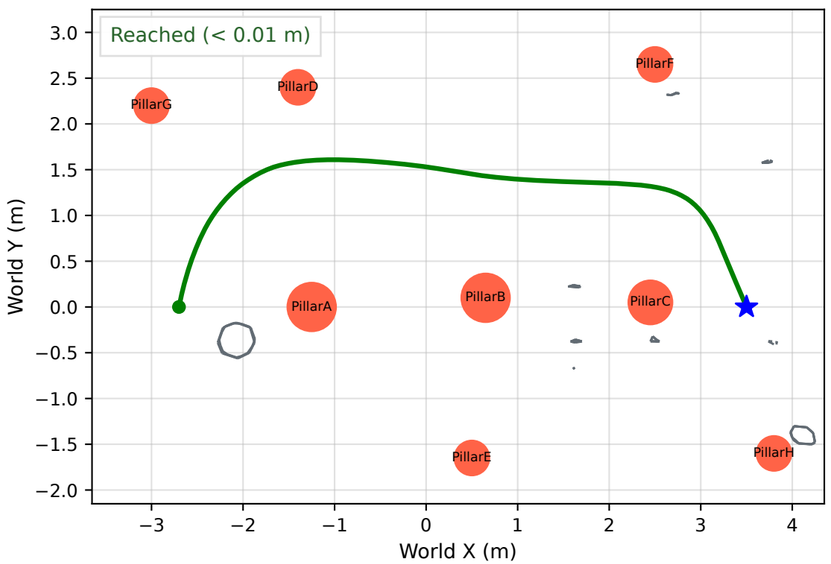
Slalom | goal error: 4.1096 m



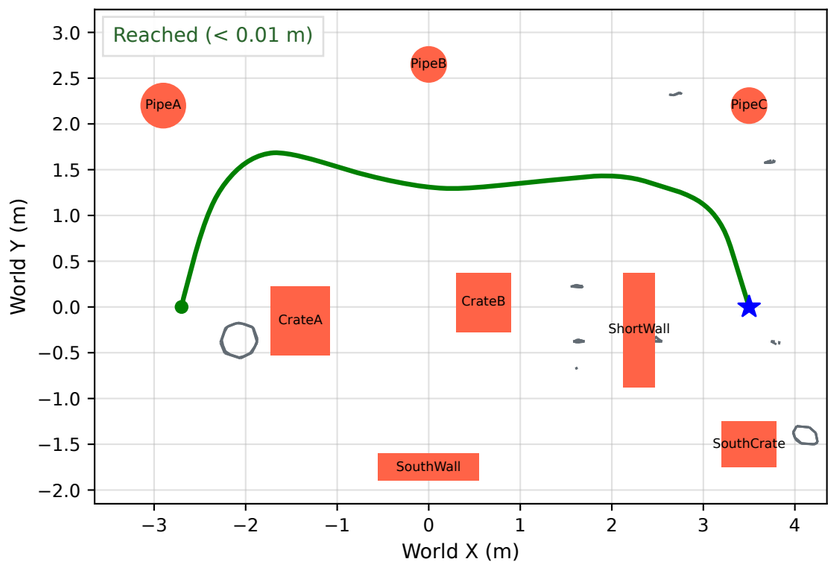
Offset gates | goal error: 2.0291 m



Pillar array | goal error: 0.0033 m



Mixed obstacles | goal error: 0.0025 m



Executed ROV path
Start
Goal
Added obstacle footprint
Original pool structures
Final position: not reached

Measured trajectories from one test suite

Fixed reference height: z = 1.0 m
Ideal depth + ground-truth odometry
Arrival tolerance: 0.01 m
Settling time: 2 s
Simulation budget: 180 s per scenario

Routes are plotted without smoothing
or extending them to the goal.